UNIVERSITY OF HEALTH AND ALLIED SCIENCES

FRED N. BINKA SCHOOL OF PUBLIC HEALTH, HOHOE

DETERMINANTS OF MOTORCYCLE CRASH INVOLVEMENT AMONG COMMERCIAL RIDERS IN SELECTED MUNICIPALITIES OF GHANA

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DECLARATION

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ABSTRACT

Background: Motorcycle crashes impose a significant public health burden in Ghana, with commercial riders among the most vulnerable road users. Despite this, evidence on crash determinants in Ghanaian municipalities remains limited.

Objectives: This study determined the prevalence of motorcycle crash involvement and identified associated demographic, behavioural, and knowledge-related factors among riders in four selected municipalities of Ghana.

Methods: A descriptive cross-sectional study was conducted among 422 systematically sampled motorcycle riders in Ablekuma Central, Yilo Krobo, Hohoe, and East Gonja. Data were collected using a structured questionnaire via KoBoCollect and analysed with descriptive statistics and multivariable logistic regression in SPSS version 26.

Results: The overall crash prevalence was 22.8%, highest in Ablekuma Central (42.7%) and lowest in Yilo Krobo (8.3%). Most riders (79.9%) lacked a valid licence, and only 50.5% demonstrated high road sign knowledge. Stunt riding(aOR=1.07, 95% CI: 1.02–1.11), riding in East Gonja versus Ablekuma Central (aOR=2.46, 95% CI: 1.06–5.71), and Muslim religious affiliation (aOR=1.77, 95% CI: 1.01–3.09) were significant predictors of crash involvement. Riders in Yilo Krobo had markedly lower odds (aOR=0.15, 95% CI: 0.05–0.51).

Conclusion: Crash prevalence was high among commercial riders, driven by risky riding behaviour and geographic location. Targeted road safety education, stricter licensing enforcement, and municipality-specific interventions are recommended to reduce motorcycle-related crashes in Ghana.

Keywords: motorcycle crashes, road safety, commercial riders, riding behaviour, road sign comprehension, Ghana

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LIST OF ABBREVIATIONS

aOR Adjusted Odds Ratio

CHAPS Community-based Health Planning and Services

CI Confidence Interval

cOR Crude Odds Ratio

DVLA Driver and Vehicle Licensing Authority

GHS Ghana Health Service

GNA Ghana News Agency

GRSF Global Road Safety Facility

LMICs Low- and Middle-income Countries

MTTD Motor Traffic and Transport Department

NRSA National Road Safety Authority

RTI Road Traffic Injury

SD Standard Deviation

SDG Sustainable Development Goal

SPSS Statistical Package for the Social Sciences

UHAS University of Health and Allied Sciences

WHO World Health Organization

CHAPTER ONE

1.0 INTRODUCTION

1.1 Background

Road traffic injuries (RTIs) remain a major global public-health and development challenge (Ahmed et al., 2023). According to the World Health Organization (WHO, 2023), about 1.19 million people died as a result of road traffic crashes in 2021, with two-thirds of those deaths occurring among adults of working age (18–59 years). Low- and middle-income countries (LMICs) bear a disproportionate share of this burden; more than 90% of road-traffic deaths occur in LMICs despite these countries owning only a relatively small share of the world's motor vehicle fleet (WHO, 2023).

Within the African Region, the challenge is particularly acute. The region recorded an estimated 225,482 road-traffic deaths in 2021, accounting for nearly 19% of global RTI deaths despite having only about 15% of the world's population and 3% of the world's vehicles (World Health Organization [WHO] African Region, 2024). Indeed, Africa holds the highest fatality rate in the world at 19.4 deaths per 100,000 population (Road Traffic Management Corporation [RTMC], 2023). From 2010 to 2021, the African region experienced a 17% increase in road-traffic fatalities contrary to the global trend of a slight decrease (WHO African Region, 2024).

Motorcycles (two- and three-wheelers) represent a particularly vulnerable mode of transport. On average, about 22.5% of road-traffic deaths in Africa in 2016 involved riders or passengers of two- and three-wheelers (Global Road Safety Facility [GRSF], 2023). In many sub-Saharan African countries, motorcyclists account for more than half of road deaths, reflecting the rapid growth of motorcycle use that often outpaces safety regulations and infrastructure improvements (FIA Foundation, 2023).

In Ghana, motorcycles often used for both personal and commercial transport are contributing significantly to the road-traffic accident burden. The occurrence of road traffic accidents (RTA) and severe crash injuries involving commercial riders has risen significantly in recent times (Konlan et al., 2022). Among persons injured from road traffic crashes in Ghana, 26% of injuries were related to motorcycles (Aidoo et al., 2021). Another study found that motorcycle crashes killed 1,056 persons and injured 4,684 commuters in 2020 in Ghana (Adanu et al., 2022). In northern Ghana, the economic burden of motorcycle accidents was estimated at US $1.2 million, of which 48% was casualty-related costs (medical costs, lost labour output, funeral expenses) and 52% was accident-related (property damage, administrative costs) (Mock et al., 2012).

Despite the heavy burden, the focus of many studies has been on overall road traffic accidents rather than on the predisposing risk factors specific to motorcycle riders. Predisposing risk factors refer to variables that increase the likelihood of a crash occurring; human or behavioural (e.g., speeding, alcohol use, helmet non-use), environmental or infrastructural (e.g., poor road condition, low visibility), vehicle-related (e.g., maintenance condition, age of motorcycle), and socio-demographic (e.g., age, experience, income) (Owusu-Ansah et al., 2024). In the Ghanaian context, non-use of helmets and male gender have been found as significant risk factors for head injuries among motorcycle-crash survivors in northern Ghana (Alhassan et al., 2022). Other studies also show that riders' attitudes and behaviours, such as intention to use helmets and perceived susceptibility, are correlated with road-safety behaviours in Ghana (Owusu et al., 2024). Riding without a helmet significantly increases the risk of fatalities and severe injuries among motorcyclists in Ghana. They face a higher likelihood of fatalities (3 times more likely to die) and traumatic brain injuries, including serious conditions like diffuse axonal injury and skull fractures. Numerous prior studies have also demonstrated that wearing a safety helmet can significantly lower the risk of head injuries (Zhou et al., 2024). This study,

therefore, seeks to identify and assess the predisposing risk factors associated with motorcycle accidents among riders.

1.2 Problem Statement

Motorcycle riders in Ghana are disproportionately vulnerable to road-traffic accidents, yet the factors that predispose riders to accidents are not sufficiently understood (Konlan et al., 2022). While motorcycle crash statistics highlight the scale of the problem, there is a lack of comprehensive, district-level quantitative evidence examining the interplay of behavioural, socio-demographic, vehicle maintenance, environmental/infrastructure and road-sign comprehension as predisposing risk factors. Without such knowledge, interventions remain generic rather than tailored to the specific risks faced by motorcycle riders. Furthermore, most existing studies have emphasized helmet use, injuries or crash outcomes rather than systematically assessing riders' risk-factor profiles and their understanding of road signs. This gap constrains the ability of policymakers, road-safety agencies and public-health practitioners to design effective, targeted strategies for reducing motorcycle-related crashes in Ghana and similar LMIC contexts.

1.3 Research Questions

1. What socio-demographic characteristics of motorcycle riders are associated with prior accident involvement in Yilo Krobo, Ablekuma Central, Hohoe and East Gonja districts?
2. What behavioral and human-related risk factors (e.g., speeding, alcohol use, helmet non-use) predispose motorcycle riders to accidents in the study sites?
3. What environmental/infrastructural and vehicle-condition factors (e.g., road condition, motorcycle maintenance) predispose motorcycle riders to accidents in the selected districts?

4. What is the level of road-sign comprehension among motorcycle riders in the study sites, and how does this relate to accident involvement and perceived risk factors?

5. What are the riders' personal attributions regarding causes of motorcycle accidents in these districts?

1.4 General and Specific Objectives

1.4.1 General Objective

To identify and assess the predisposing risk factors associated with motorcycle accidents among riders in selected districts of Ghana.

1.4.2 Specific Objectives

1. To assess the prevalence of motorcycle accidents across the selected districts

2. To assess the knowledge of road signs across the selected districts

3. To identify the Demographic and behavioural factors associated with motorcycle accidents in the selected districts.

1.5 Significance of the Study

This study is significant in several respects. First, it will generate new quantitative evidence on predisposing risk factors for motorcycle accidents across multiple districts in Ghana, thereby filling a gap in the literature. Second, by focusing on riders' comprehension of road signs alongside behavioural, vehicle and environmental factors, the study brings a novel dimension to motorcycle-safety research in Ghana, which has so far concentrated largely on injury outcomes and helmet use. Third, the findings will assist road-safety agencies (e.g., the National Road Safety Authority), traffic regulators, transport-policy makers and rider associations to design targeted interventions (such as training modules, enforcement campaigns, infrastructure improvements and signage education) tailored to risk-profiles identified in both urban and rural

settings. Finally, as motorcycle use continues to expand in Ghana and across sub-Saharan Africa, the evidence from this study may have applicability beyond the study sites contributing to broader regional efforts to achieve the United Nations Sustainable Development Goal 3.6 (halving global road-traffic deaths by 2030) and SDG 11.2 (safe, affordable transport systems).

1.6 Scope of the Study

This study is structured to focus on motorcycle riders (commercial and private) operating in four Ghanaian districts: Yilo Krobo Municipal (Eastern Region), Ablekuma Central Municipal (Greater Accra Region), Hohoe municipality (Volta Region) and East Gonja Municipal (Savanna Region). It employs a quantitative survey design using a structured questionnaire to assess prior accident involvement, behavioural/human factors, vehicle and environmental/infrastructure factors, and riders' comprehension of a selected set of eight road signs. The study will not include other road-user categories (e.g., pedestrians, bus drivers) nor will it directly investigate post-accident injury outcomes or long-term disability. The data collection will be cross-sectional and limited to riders available during the study period.

CHAPTER TWO

2.0 LITERATURE REVIEW

2.1 Introduction to the literature review

This chapter synthesizes empirical and conceptual literature relevant to the predisposing risk factors of motorcycle accidents. It is structured to address the study's specific objectives: (1) the prevalence of motorcycle accidents, (2) the knowledge of road sign, and (3) Demographic and behavioural factors associated with motorcycle accidents. Each section reviews global, regional (Africa), and Ghanaian evidence to situate the current study in the local and international context.

2.2.1 Prevalence of motorcycle accidents

Motor-cycle accidents comprise a greater and, compared to other road-traffic injuries (RTIs), disproportionately high proportion of road traffic casualties in Ghana, especially between commercial riders (localized as okada operators) who comprise a medium of unregulated but central constituency in Ghanaian urban and peri-urban municipal areas. As stated by the National Road Safety Commission (NRSC) of Ghana, in 2020 alone, the crashes among motorcycles have resulted in 1,056 fatalities and 4,684 injuries (referenced in Marnah and Manortey, 2022). These figures demonstrate the growing burden on community health, whereby the motorcycle crashes have increased to 21.75 per cent between 4, 643 incidents in 2019 and 5,684 cases in 2020. As one of the most vulnerable types of road users, motorcyclists in the country are estimated to be the leading cause of all the road traffic incidences occurring nationwide with an estimated percentage of nearly 34, nonetheless, being higher in the municipalities where commercial motorcycles activities have offered a significant source of livelihood to the unemployed youth. The trend is synonymous to other Sub-Saharan African

trends, whereby lack of formal training, licenses and formal training of riders exposes and crashes among commercial operators.

Hospital-based evidence is also a measure of the incidence of motorcycle accidents that are localised to the healthcare system of Ghana. Using Cochran formula, Marnah and Manortey (2022) strategically selected 200 motorcycle accident victims who had been taken to the hospital (Korle-Bu Teaching Hospital (KBTH)) and were aged 18 years or above as the main national referral centre in the country when it comes to motorcycle accidents. The calculation of the amount of the sample was founded on a literatures prevalence rate of 37 percent of motorcycle accidents in the past (Luther, 2019), as adjusted to the levels of a 95 percent confidence interval, a 5 percent margin of error, and a 10 percent non-response rate. Socio-demographic profiling of these victims showed that commercial riders were evidently given the center stage of involvement in the crash: approximately a third of the victims were aged at 18 years of age, 99% were men, 56.5% were single; 68.5% were Christians; 33% had only made it through junior high school education. It is worth noting that 24.5 per cent of the subjects said that their prime occupation was commercial motor riding despite the majority (84.5) of the respondents saying [?]15 years' experience in motor riding. The number of licensing was alarmingly low where 40 per cent had no valid rider licence and 27 per cent had expired licences. Not over 40 per cent of the respondents indicated that they had reasonable insurance cover in terms of motor or national health. These characteristics serve to prove that the rate of motorcycle accidents is not evenly spread, but rather concentrated to a large extent in the inexperienced and unlicensed commercial motorcycle riders in high exposure settings like in Greater Accra.

The sub-national estimates are allowed to support these complementary national and hospital level estimates. Luther (2019) estimated a study-wide motorcycle accidents prevalence of 37 percent in an unpublished thesis on prevalence and correlating factors in Accra that directly

informs the sampling frame in the KBTH study and identified the predictability of the large cities in supporting the existence of high crash rates. Even higher contextual data provided by Marnah and Manortey (2022) indicate that the road traffic accidents in Ghana are reported to be claiming at an average of four lives per day and that the Ashanti, Eastern, Greater Accra, central and Brong Ahafo region is also responsible of more than three-quarters of the number of national crash fatalities. These deaths are classified or categorized almost evenly among the youthful generations (16-45 age group) (58%); this group of death is essentially part of commercial riding population.

The extra indirect evidence of the driver factors behind high prevalence is provided by the behavioural research concerning the noncompliance of riders. In one of the studies on noncompliance with traffic control in the Wa Municipality on riders of motorcycles, 97.8 percent of 90 sampled motorcycles showed evidently high degrees of knowledge deficiency as regards traffic regulations, and therefore, they showed small percentages of sampled motorcycle riders (90) fully knowledgeable on traffic regulations (systemic knowledge gaps pushing this percentage across the sample). Broad scale indiscipline, absence of accreditation of the training as well as absence of licensing were cited as the factors that led to high risk of crash, trends that were similar to the licensing and experience shortfalls of KBTH cohort. The issue of commercial riders with general enforcement is not exclusive, but signatory of noncompliance in high levels of crash prevalence in commercial riders in selected municipalities.

Together, these empirically based sources are national NRSC statistics, prevalence measurement tool of the KBTH hospital (Marnah and Manortey, 2022), the benchmark in Prevaccinist Accra itself (Luther, 2019), and behavioural evidence on Wa Municipality which is used to determine that the prevalence of motor accidents in Ghana is continually high, that is, it concerns an area of commercial riding. The 34 percent contribution to national RTIs

recorded, the 2020 loss of more than 5,600 crashes and more than 1,000 deaths, and the case burden of young and non-licensed operators in urban areas on unfavorable structures testify to a community health crisis facilitated by the amount of exposed so far, absence of facilities and inadequate regulations. The prevalence patterns define the platform to ponder on specific determinations, though demographic, behavioural, environmental and infrastructural aspects to be discussed afterwards, as they require preliminary interventions to achieve the importance of the reduction of both the magnitude and trends of the issue to achieve business-level rider crash participation intercessions.

2.2.2 Knowledge of road signs

In a study conducted by Marnah and Manortey (2022) to investigate the prevalence of the motorcycle accidents that were brought to the Korle-Bu Teaching Hospital in Ghana showed that the majority (78.05 percent) of sampled contestants were aware about road laws prior to the accident. The authors note that the big sources of this information were primarily non-formal with the attendance at workshops having the smallest percentage with the biggest fraction of the portions having their knowledge to have been acquired by other means primarily through informal media or peer networks. It concluded that the respondents were highly familiar with the Prohibitory, Mandatory and Informatory signs with only a few having a little knowledge about the Breakdown and Accident signs. Despite the view of the respondents being that they knew the prohibitory signs, following the objective assessment, 65 percent of them were not in a position of recalling or naming the prohibitory signs, even the most common ones, and gave a response of I do not know. The causes of their predicament as findings by the accident victims lie in the non-compliance with road signs and speeding, and there was a gap between the perceived compliance and awareness with the perceived awareness as reported.

Similarly, a cross sectional study conducted in the Wa Municipality, regarding the failure of traffic control lights among motorcycles riding uncovered that there was a huge ignorance of road regulations among commercial riders. The analysis has found out that significantly a majority of the riders were found to have dramatic levels of deficiency in regards to the knowledge of the rules, the structural factors were also included and they were lack of instructor certification and professional licensing. The reason why some of the factors that facilitated the low level of knowledge was due to poor implementation of road regulations and failure to offer rider training programmes.

In the second study, on awareness of road regulation among the motorcycle riders in the Bolgatanga Municipality, it was established that the awareness levels in the sample were also very different. The researchers have found out that the demographic factors were highly significant and that dissimilarities were observed among the subjects in regard to the familiarity with the traffic signs and laws. Factors that created such discrepancies include degree of education and age of the riders.

A similar assessment of the level of knowledge and awareness of road signs and rules among the adults of the sampled Ghanaian settings revealed that there was high self-reported basic knowledge of the road signs but little basic practice. Majority of the respondents who had heard about road rules showed superficial understanding of road rules with particular regard to ideas of signs during the state of riding in reality. The level of knowledge on road signs has been discovered to be associated with education and exposure to formal training.

In assessing the knowledge of the road signs and knowledge of the road signs and traffic rules among the motorcyclists in LEKMA Hospital catchment area, the research found out that the most recognised category of road signs was the prohibitory sign among the motorcyclists. General knowledge that would guarantee a safe ride was still inadequate and majority of riders could not properly use the signs when engaging in some risky motorcycle maneuvers such as

passing over another bike or making a cross. The study also established that not following the traffic laws had a direct relationship with the absence of formal education.

As Ngim and Udozen (2007) in their research on factors that determine adherence to road traffic regulation in Ghana contends, they lack the functional knowledge on road signs and road regulation. Part of the factors that contributed to the poor knowledge of the riders includes the education level, the absence of structured training programs, informal nature of information used.

In another study to establish knowledge and perception of road safety among road signs among commercial motorcycle riders in urban Ghana, it was established that 78.0 percent of the motorcyclists reported high knowledge on road signs with majority of them having earned the knowledge information in non-traditional media sources. However, the researchers found out that higher proportion of all riders shared more than ten years of riding experience with few few subjects reporting flawless understanding of road signs. Life in the city municipalities, marriage and profession were also positively correlated with good knowledge of road signs and rules.

Combined, these articles drive home the case that, despite the high self-reported road sign knowledge between commercial motorcycle riders in Ghana in the areas of the hospital and a community-based sample quite high at 78 percent and 78 percent respectively in recall and application respectively, it is superficial and unsatisfactorily high. The interrelationship literature reveals that poor means of educational attainment and bureaucracies involved in the training of those who ride and use of informal channels of information has increased knowledge gaps. The above trends are the direct cause of the occurrence of non-adherence behaviours, which are observed through motorcycle crashes in the sampled municipalities and the need to have a focused and systematic road sign system education programme, in combination with

licensing and enforcement regimes, to increase compliance and participation of fewer people in the crashes in commercial motorcycle riders.

2.2.3 Demographic and behavioural factors associated with motorcycle accidents

In an article by Marnah and Manortey (2022) focusing on the prevalence and risk factor of motorcycle accidents in the Korle-Bu Teaching Hospital in Ghana, the use of the motorcycle helmet by the affected victims is found to be significantly correlated to age, marriage status, education and occupation. The researchers assumed that the sample was relatively young (18-25 years old), and that it consisted of predominantly males, the education level was not higher than that of a junior high, and that it also included commercial motor riders whose main occupation is this trade. The number of victims themselves noted behaviour elements as the main cause of car accidents; the inability to follow road signals and over-speeding. It was even determined that the proportion of respondents that had not obtained an eligible rider licence or that their licences had expired was staggering and that less than 40 percent of respondents insured their motorcycles or had a valid national health insurance. The trends show that failure of behaviour, other than demographic predispositions, are cohorts to augment rate of involvement of commercial riders in crashes in urban municipalities.

Likewise, within a sample of the commercial motor-bike riders in Ghana, the cross-sectional research study carried out showed that, demographic factors, which lead to a better engagement in the crash include; low educational status, a single marital status, work occupation of commercial riding besides young age. Among the behavioural threats tested by the researchers are untrained behaviour, licensing that is not real, deficit of riding experience and utilisation of unsanctioned peer groups to extract data about road safety, concepts, which contribute a great deal in the rise of accidents. Some of the reasons that have contributed to such lapses in behavioural practices have included: No official programmes in training the riders and leniency

in following through with the provisions of licensing i.e. most okada operators are not educated about the consequences of the actions of confrontation causing traffic in the city.

The other investigation study that was performed on the topic of accidental traumas and fatalities as the second under-investigated infection in the entire world of road traffic accidents displayed that the youth and the male gender is the cause of the motor-cycle crashes in the universe where Ghana has shown the greatest influences in the low- and middle-income nations (Ahmed et al., 2023). The authors have come to a conclusion that common behavioural criteria leading to increased severity of injury involve helmet off application, risk-taking through aggressive overtaking and exhaustion in the case of two-wheel cars. Some of the agents, which are thought to cause these risks, include need to meet commercial transport demands, lack of road safety and lack of formal training in safety related field.

Under the framework of demographic and behavioural variables depicted by the population in regards to the motorcycle accidents of commercial riders in Africa, a intended scoping review of the nonmedical use of tramadol by Boun et al. (2024), denotes that commercial motorcyclists are being given prodigal amounts of nonmedical use of tramadol of 76 percent. With the help of exhaustion as a lever to eliminate getting up and the continuation of the ride after having passed the customary milestones, the researchers found this form of substance abuse, to be a life threatening behaviour with its risks being quite significant, and in the vast majority of cases resulting in a significant enhancement of the degree of judgment, the speed of reactive and decision making. Adults, active in their job, younger adults are the population groups which have been detected to exert by the greatest effect to which uncontrollable use of such substances and occupation in the transport workers was established to be one of the dominant factors that pose a threat to the transport workers. The same review also found that that also dependence and withdrawal symptoms related to the tramadol action predisposes to risky ride behaviour, a

vicious circle of increasingly serious crashing incidents among commercial operators existed, in the municipalities of urban and peri urban nature.

The demographic predictors of low compliance regarding the use of helmets, not complying with road signs and wearing helmets at over speed in the parallel comparison of the level of quantity knowledge, awareness and practices of road safety among motorcyclists in Ghana are level of education and youthfulness respectively. It was observed that that most of the commercial riders used non-official sources of information instead of official sources of training, so did not get a lot of information on safety measures. Part of these causes will include, a want of regular, repeated mass-education, of the transport workers of the municipality, etc. in particular, the economic ones which carry the unemployed youths to okada business with no semblance of preparation whatever, etc.

A combination of these findings implies that the youth, being a man, being poorly educated, being a single man having commercial riding vocation, and being a bad rider (apparently, the respondents were observed not to have seen road signs) were the factors that enhanced the experience of motorcycle crashes among the sampled commercial riders in the municipalities in Ghana. The areas are erroneously described as traditionally enhanced through a network wide deficiency in formal education, law and occupation. To fight them, they should be diagnosed together to manipulate the demographic characteristics of the population under risk and modify their behaviour that can be manipulated to maintain the high crash rates to provide them with more favourable treatment by road than many other problems that the national and international priorities take into consideration.

CHAPTER THREE

3.0 METHODS

3.1 Introduction

This section presented a comprehensive description of the methods and techniques that were used in carrying out the study. These included the research design, study site, population, sample size, sampling techniques, research instruments, data collection, and methods which were used to analyze the data.

3.2 Study Design

This study employed a cross-sectional descriptive design using a quantitative approach to examine the predisposing risk factors associated with motorcycle accident involvement among riders. The cross-sectional design was appropriate because it allowed data to be collected from a sample of respondents at a single point in time to describe and analyze the relationships between variables such as socio-demographic characteristics, environmental and vehicular conditions, road-sign comprehension, and riders' attributions of accident causes (Creswell & Creswell, 2018).

3.3 Study Sites

The study was conducted in four selected districts across Ghana, chosen to represent a mix of urban, peri-urban, and rural settings and to capture diversity in infrastructure, rider profiles, and accident contexts. These were Yilo Krobo Municipal (Eastern Region), Ablekuma Central Municipal (Greater Accra Region), Hohoe Municipal (Volta Region), and East Gonja Municipal (Savannah Region).

3.3.1 Yilo Krobo Municipal (Eastern Region)

Yilo Krobo Municipal is located in the Eastern Region of Ghana, with Somanya as its capital. The terrain is hilly, and motorcycles are widely used for both personal transport and commercial passenger riding due to the scarcity of public transport in remote communities. The district has reported fatal collisions involving motorcycles and issues such as dust and poor visibility on unpaved roads (Ghana News Agency, 2024). For example, a crash on the Huhunya Asesewa road in Yilo Krobo resulted in one death and 14 injuries, attributed in part to dusty road conditions and poor visibility (MyJoyOnline, 2024). Although there is no official published count of registered motorcycle riders in the Municipality, the reliance on motorcycles as a major mode of transport and frequent road crashes suggest a substantial rider population. The study site thus offers a peri-urban and semi-rural environment with moderate road infrastructure, making it suitable for exploring predisposing risk factors in transitional contexts.

3.3.2 Ablekuma Central Municipal (Greater Accra Region)

Ablekuma Central Municipal is one of the urban districts in the Greater Accra Region, characterised by high population density, congested roads, and significant commercial motorcycle ("okada") activity. According to national road safety statistics, motorcyclists and pedestrians represent the largest share of traffic-related fatalities in the Greater Accra and Ashanti regions (The Ghana Report, 2024). Additionally, official reports indicate that the number of registered motorcycles in Ghana increased sharply between 2010 and 2021, with the Greater Accra Region hosting one of the highest concentrations of motorcycles (Graphic Online, 2024). Although specific data for Ablekuma Central are not published, the urban context implies a large commercial rider population exposed to dense traffic, short trip patterns, and frequent interactions with other vehicles, key situational risk factors for motorcycle crashes.

3.3.3 Hohoe Municipal (Volta Region)

Hohoe Municipal is located in the Volta Region, which has shown elevated motorcycle-related accident rates and a high dependence on motorcycles for transport due to its terrain and limited access to public transport. A cross-sectional study in Hohoe involving 238 commercial motorcycle riders reported that only 59.2% complied with safety regulations (Hagan et al., 2021). The district population was about 144,502 according to the 2000 Population and Housing Census, with an annual growth rate of 1.9% (Ministry of Food and Agriculture [MoFA], 2024). Although no official record exists on the total number of registered motorcycle riders, the existence of rider-focused studies and the prevalence of commercial transport suggest a large rider population. Hohoe therefore provides a mixed urban-rural context that enables examination of behavioral and infrastructural determinants of motorcycle accidents in a semi-urban setting.

3.3.4 East Gonja Municipal (Savannah Region)

East Gonja Municipal (Salaga), located in the Savannah Region, is predominantly rural and agrarian, with motorcycles serving as the main mode of passenger and goods transport. The district population was approximately 117,755 in 2021 (Wikipedia, 2024). Road infrastructure is less developed compared to urban regions, enforcement is limited, and riders often cover long distances between towns, increasing exposure to road hazards. Although the number of registered motorcycles is undocumented, the frequent use of motorcycles and reports of crashes in the area make it an important site for investigating environmental and human-related risk factors in rural contexts.

3.4 Study Population

The study population comprised registered and unregistered commercial motorcycle riders and personal users operating within the four selected municipalities in Ghana: Yilo Krobo Municipal (Eastern Region), Ablekuma Central Municipal (Greater Accra Region), Hohoe

Municipality (Volta Region), and East Gonja Municipal (Savannah Region). These individuals represented both formal and informal riders who used motorcycles for passenger transport ("okada") or goods delivery, or noncommercial users within their respective areas.

In addition to motorcycle riders, key informants such as officers from the Motor Traffic and Transport Department (MTTD), and other municipal and community transport executives, popularly referred to as "station masters," were included to provide information on the number of transport units as well as the number of registered riders within each district.

3.4.1 Inclusion Criteria

- Commercial motorcycle riders (both self-employed and employed riders) who have operated for at least six months within the study municipality.
- Noncommercial motorcycle riders who have been actively riding for at least six months
- Riders aged 18 years and above.

3.4.2 Exclusion Criteria

- Riders who are merely passing through the study sites (Yilo Krobo, Ablekuma Central, Hohoe, and East Gonja) but do not habitually operate or belong to a local commercial "station" or "junction" within these districts.
- Riders who are physically or mentally unfit to participate at the time of data collection (e.g., due to severe injury or illness).
- Riders with severe hearing or speech impairments that would significantly hinder the administration of the questionnaire or the assessment of road-sign comprehension.

3.5 Sample size determination (cross-sectional survey)

1) Choice of formula and assumptions

The standard Cochran formula was used for estimating a sample size for a proportion in a large population:

$$n_0 = \frac{Z^2\, p(1-p)}{d^2}$$

Where:

- Z= 1.96 (for 95% confidence)
- p= assumed proportion (prevalence). Because there is no precise prevalence to base this on for "ever had an accident" across the four districts, we use $p = 0.5$(conservative choice that maximizes required sample size).
- d= desired margin of error = 0.05 (5%).

Step-by-step arithmetic:

1. $Z^2 = 1.96 \times 1.96 = 3.8416.$
2. $p(1-p) = 0.5 \times 0.5 = 0.25.$
3. Numerator $= 3.8416 \times 0.25 = 0.9604.$
4. Denominator $d^2 = 0.05 \times 0.05 = 0.0025.$
5. $n_0 = 0.9604/0.0025 = 384.16$`

2) Adjustment for non-response / contingency

The study allows for a 10% non-response or unusable questionnaires:

$$n_{\text{adj}} = n_0 \times (1 + 0.10) = 384 \times 1.10 = 422.4$$

3) Allocation of total sample across the four study sites

Because reliable counts of registered motorcycle riders per district are not consistently available online, the study used the 2021 Population and Housing Census (Ghana Statistical Service) district populations as the best proxy to allocate sampling effort proportionally. The 2021 district populations are:

- Yilo Krobo Municipal = 122,705. (GSS 2021)
- Ablekuma Central Municipal = 169,145. (GSS 2021)
- Hohoe Municipal = 114,472. (GSS 2021)
- East Gonja Municipal = 117,755. (GSS 2021)

  Total population across the four districts = $122{,}705 + 169{,}145 + 114{,}472 + 117{,}755 = 524{,}077$.

  The study allocates proportionally:

$$\text{Sample}_i = n_{\text{adj}} \times \frac{\text{Population}_i}{\text{Total population}}$$

- Yilo Krobo: $422 \times \left(\frac{122{,}705}{524{,}077}\right) = 422 \times 0.2342 = 98.8$ =99 respondents.
- Ablekuma Central: $422 \times \left(\frac{169{,}145}{524{,}077}\right) = 422 \times 0.3229 = 136.2$ =136 respondents.
- Hohoe: $422 \times \left(\frac{114{,}472}{524{,}077}\right) = 422 \times 0.2185 = 92.2$ =92 respondents.
- East Gonja: $422 \times \left(\frac{117{,}755}{524{,}077}\right) = 422 \times 0.2244 = 94.8$ =95 respondents.

  Sum: $99 + 136 + 92 + 95 = \mathbf{422}$

3.6 Sampling technique and procedure

A multistage stratified cluster sampling technique was employed. This approach was widely used in transportation and public health research where target populations were mobile and naturally organized into operational clusters such as motorcycle ranks, stations, or major junctions (Levy & Lemeshow, 2013; Cochran, 1977).

3.6.1 Sampling Technique

The sampling followed a multistage process involving stratification, cluster selection, and systematic sampling of individual riders within clusters. Stratification was based on the four study districts: Yilo Krobo Municipal (Eastern Region), Ablekuma Central Municipal (Greater Accra Region), Hohoe Municipal (Volta Region), and East Gonja Municipal (Savannah Region) to ensure geographical representation. Within each district, clusters were defined as motorcycle stations, popular loading points, or major junctions where riders usually assembled. A probability-proportional-to-size (PPS) approach was used to select clusters when estimates of rider numbers per station were available. In the absence of accurate counts, simple random sampling of clusters was applied. Within each selected cluster, systematic sampling with a random start was used to select respondents. This approach minimized interviewer bias and provided each rider with a known, non-zero probability of selection.

3.6.2 Sampling procedure

The total sample of 422 riders was proportionally allocated across the four districts based on population size or available rider estimates. Tentatively, the distribution was: Yilo Krobo (99), Ablekuma Central (136), Hohoe (92), and East Gonja (95).

In each district, a list of active motorcycle ranks, loading points, and major traffic junctions was compiled in consultation with the Motor Traffic and Transport Department (MTTD), local rider unions, and district assemblies. Between six to eight clusters were selected per district depending on rider density and coverage needs (Dapilah et al., 2017).

Where cluster size estimates were available, clusters were selected using probability-proportional-to-size (PPS) to ensure that larger stations had a higher chance of inclusion. Where this was not feasible, simple random sampling of mapped clusters was used.

On the survey day, enumerators estimated the total number of eligible riders present at the cluster (N). The required number of riders to be interviewed from that cluster (n) was predetermined based on district quota allocation. The sampling interval (k) was calculated as. A random start between 1 and k was chosen, and every k^{th} rider was approached for the interview until the cluster quota was reached. Refusals or ineligible participants were replaced by the next eligible rider following the same systematic pattern.

Because private riders might not have congregated at ranks, additional sampling points such as fuel stations, market areas, and roadside junctions were included using time-location sampling to ensure representation of both commercial and private riders.

3.7 Data Collection Procedure

Data were collected through structured interviewer-administered questionnaires using KoBoCollect. Data on commercial motorcycle riders from various districts, focusing on socio-demographic characteristics, risk factors such as speeding and helmet use, environmental factors like road conditions, and riders' understanding of road signs and previous accidents. Data collection involved both face-to-face and online interviews conducted by a fluent researcher, ensuring effective communication. The process lasted for weeks, with daily reviews for data quality and consistency.

3.7.1 Development of the Data Collection Instrument

The questionnaire was designed based on literature on motorcycle safety, risk perception, and behavioural attributions (Dapilah et al., 2017; Hagan et al., 2021; Owusu et al., 2024). It will be structured into six major sections:

Socio-demographic characteristics: Age, sex, marital status, education, income, riding experience, and licensing status.

Riding behaviours and safety practices: Helmet use, speeding, alcohol consumption, phone use while riding, and adherence to traffic rules.

Compliance with traffic regulations: Possession of a valid driving licence, registration, insurance, and awareness of the Road Traffic Act.

Crash history and exposure: Previous involvement in accidents, injury outcomes, and perceived causes of crashes.

Knowledge and comprehension of road signs: This section assessed participants' ability to recognize and interpret selected Ghanaian road signs. Ten representative signs (e.g., "no overtaking," "speed limit," "pedestrian crossing," "sharp bend ahead") were displayed as images within the KoBoCollect form. Respondents identified each sign and explained its meaning.

Causal attributions for accidents: Participants were asked to indicate what they believed were the most common causes of motorcycle crashes (e.g., self-error, poor roads, weather, mechanical failure, fate, or other drivers).

The instrument was pre-tested among 20 commercial riders in a neighbouring municipality (Kpando) to ensure clarity, logical flow, and ease of navigation on the mobile device. Feedback from the pilot was used to refine question wording, add local examples, and adjust skip patterns before the actual data collection.

3.7.2 Data Collection Process

Data collection took place over two weeks, using trained enumerators familiar with public health fieldwork and road safety studies. Enumerators spoke English and relevant local languages (Ewe, Twi, Dagbani, Dangme) to ensure clear communication. Each enumerator was equipped with an Android tablet preloaded with the final KoBoCollect questionnaire.

Data were collected offline and synchronized daily to the secure KoBoToolbox server when internet connectivity was available. GPS tracking was enabled to verify field locations and prevent data falsification. During each interview, enumerators explained the study objectives, obtained digital informed consent, and administered the survey in a private, comfortable space to ensure confidentiality.

3.8 Data Analysis

Data were exported from KoBoCollect in Excel format and cleaned before being analyzed using the Statistical Package for the Social Sciences (SPSS) version 26.0. The analysis proceeded in several stages: data cleaning and coding, descriptive analysis, bivariate analysis, and multivariate modeling to comprehensively address all study objectives.

3.8.1 Data Cleaning and Coding

Collected data were reviewed for completeness, consistency, and accuracy. Any duplicate or incomplete entries were excluded. Categorical variables such as gender, marital status, helmet use, and accident history were coded numerically (e.g., 1 = Yes, 0 = No). For the road-sign comprehension test, responses were scored as *correct (2 points)*, *partially correct (1 point)*, or *incorrect (0 points)*, with a composite score computed to assess overall knowledge level. Scores were then categorized as *good*, *moderate*, or *poor* comprehension based on percentile cut-offs.

Descriptive Analysis

Descriptive statistics summarized participants' socio-demographic characteristics and key variables. Frequencies, percentages, means, and standard deviations were computed for categorical and continuous variables respectively. Graphical representations such as bar charts and pie charts were used to display the distribution of responses (e.g., helmet use, crash involvement, and road-sign knowledge levels).

Bivariate Analysis

Bivariate analysis were performed to examine associations between selected independent variables (e.g., age, education, experience, helmet use, alcohol use, road-sign knowledge, and vehicle condition) and motorcycle accident involvement (dependent variable).

Chi-square tests (χ^2) were used to determine significant associations between categorical variables.

Independent t-tests were used for comparisons of mean road-sign knowledge scores between groups (e.g., riders with vs. without crash history). Variables that show statistical significance at $p < 0.05$ in bivariate analyses were included in the multivariate model.

Multivariate Analysis

Binary logistic regression was employed to identify independent predictors of motorcycle accident involvement while controlling for potential confounders. Adjusted odds ratios (AORs) with 95% confidence intervals (CIs) will be reported. Potential predictor variables will include: Socio-demographic characteristics (age, gender, education level, years of riding experience) Behavioural factors (speeding, helmet use, alcohol use, mobile phone use while riding) Environmental/vehicle factors (road condition perception, motorcycle maintenance practices) Knowledge and comprehension of road signs Causal attributions for accidents

Subgroup and Comparative Analyses

Comparative analyses were conducted across the four study municipalities (Yilo Krobo, Ablekuma Central, Hohoe, and East Gonja) to explore spatial variations in risk factors and safety practices. An ANOVA test will be used to assess differences in mean scores (e.g., road-sign knowledge) between districts. Crosstabulations and chi-square tests will further examine district-level variations in accident history and compliance rates.

Statistical Significance and Presentation

A **95% confidence level** were used for all analyses, and statistical significance will be set at $p < 0.05$. Results were be presented in tables and charts, accompanied by narrative interpretation.

3.9 Ethical Issues

The UHAS Research Ethics Committee were consulted for ethical approval (UHAS-REC).

3.10 Compensation

Participants were not compensated for their participation. No expenses were made as a result of participation. However, participants were duly acknowledged in the project as an appreciation for their time and participation in the study.

3.11 Data Storage and Ownership

Data were stored by the researchers. The data were made accessible to members of the research team for analysis purposes, and to the authorities of the School of Public Health, University of Health and Allied Sciences.

3.12 Voluntary Participation and Withdrawal

Participants were informed that participating in this study is voluntary and therefore, they have the right to participate or refuse to participate in the study. Withdrawal from the study at any point in time is without any consequences.

3.13 Informed Consent Process

All the study participants had the right to ask questions concerning the study or their participation in the study. Answers were readily provided to clarify doubts about participation and other issues concerning the study by the researcher and or her supervisor. After participants agreed to participate in the study, written informed consent were obtained from each one of them.

3.14 Declaration of Conflict of Interest and Funding

No sponsorship was secured for the study. The study was solely be funded by the researchers.

3.15 Confidentiality and privacy

Before each participant participates in the interviews, the study's aims were explicitly communicated to them, and their informed consent were asked. The respondent was assured that his/her answers will remain confidential and cannot be traced back to them in any way as we would not take any personal information such as name, house address, etc.

3.16 Study Involvement and Withdrawal

Participants were notified that their participation in the study is completely optional and that they could opt-out at any time during the study. Before being questioned, participants who only consent to participate in the study will be given a consent form to sign/thumbprint.

CHAPTER FOUR

4.0 RESULTS AND DISCUSSION

4.1.1 Sociodemographic characteristics

Table 1 presents the socio-demographic and riding characteristics of respondents. The mean age of respondents was 33.58 ± 10.38 years, with the majority (44.3%) aged 25–34 years. Most respondents were male (88.4%), while females accounted for 11.6%. Slightly over half of the respondents were Christians (50.9%). Regarding education, 36.5% had secondary education, while 35.6% had tertiary education. In terms of riding experience, 46.7% had been riding for more than five years, and most respondents rode motorcycles every day (65.6%). Additionally, 59.5% used motorcycles for non-commercial purposes, while 79.9% reported not having a motorcycle license. The majority (32%) of respondents lived in the Ablekuma Central district, while 60.7% of respondents owned a motorcycle.

Table 1 Sociodemographic characteristics

Variables	**Frequency (n=422)**	**Percentage (%)**
Mean (±SD)	± 33.58 (10.38)	
Age group		
Below 25	73	17.3
25-34	187	44.3
35-44	88	20.8
45-54	58	13.7
55+	16	3.7
Sex		
Female	49	11.6

Male	373	88.4
Religion		
Christian	215	50.9
Islam	197	46.7
African Traditionalist	10	2.4
Marital Status		
Not in union	248	58.8
In union	174	41.2
Educational attainment		
No formal education	50	11.8
Basic education	68	16.1
Secondary education	154	36.5
Tertiary education	150	35.6
District		
Ablekuma central	135	32.0
Yilo Krobo	99	23.5
East Gonja	95	22.5
Hohoe	93	22.0
Riding years		
Less than 1	50	11.8
1-5	175	41.5
Above 5	197	46.7
Riding frequency		
Every day	277	65.6
Several times per week	119	28.2

Once a week	15	3.6
Rarely	11	2.6
Riding purpose		
Commercial	171	40.5
Non-commercial	251	59.5
Motor ownership		
Yes	256	60.7
No	166	39.3
License		
Yes	85	20.1
No	337	79.9

4.1.2 Prevalence of accidents by districts

Figure 2 shows the prevalence of motorcycle accidents across the selected regions. With regards to riders who reported being involved in motorcycle accidents, Ablekuma Central reported the highest prevalence (42.7%) of motorcycle accidents, while Yilo Krobo reported the lowest prevalence (8.3%). Concerning riders who reported no accidents, Yilo Krobo reported the highest rate (27.9%), with Hohoe recording the lowest rate (10.4%).

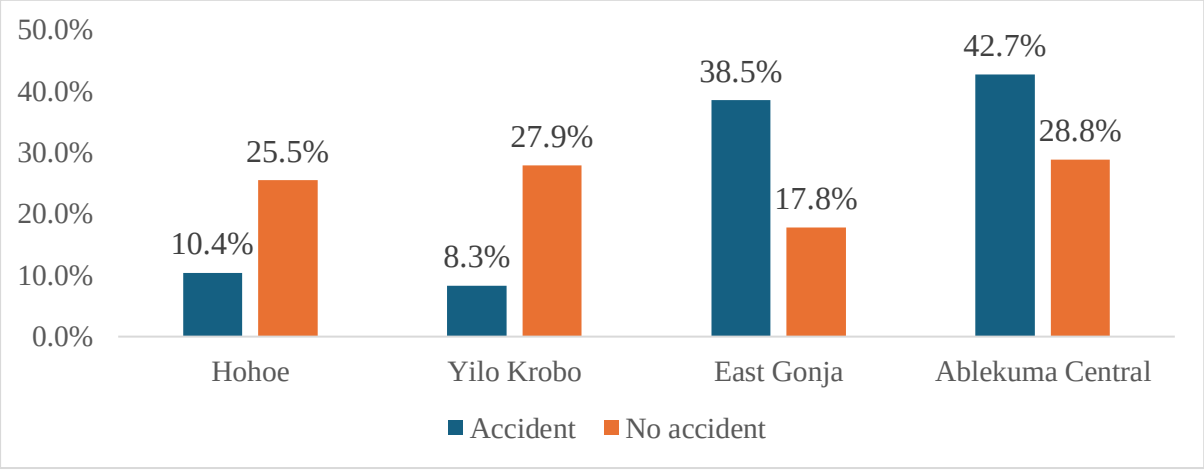

Figure 1 Prevalence of accidents by districts

4.1.3 Overall prevalence of motorcycle accidents

The results show the overall prevalence of motorcycle accidents across the selected districts. The prevalence of motorcycle accidents among respondents was 22.8% across the four study districts.

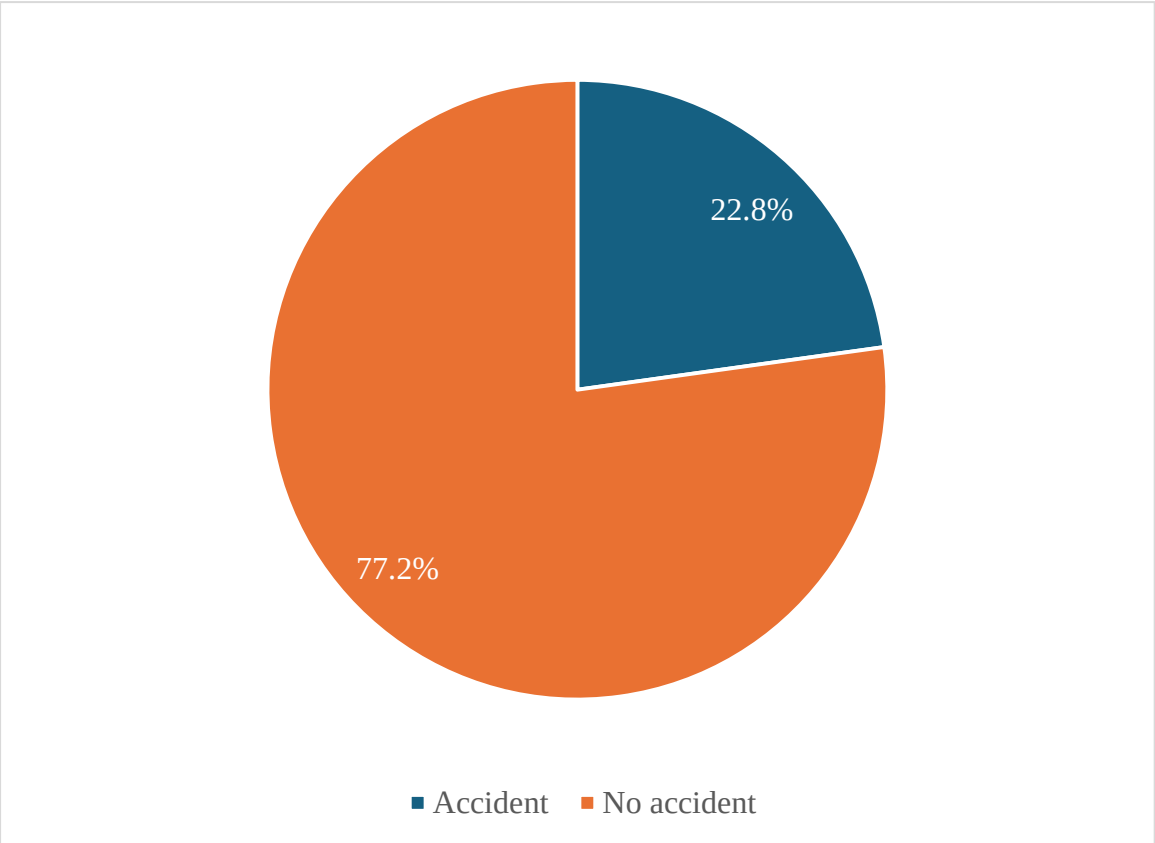

Figure 2 Overall prevalence of motorcycle accidents

4.1.4 Knowledge of road signs among respondents

The results in Table X show varying levels of knowledge of road signs among respondents. The majority of riders correctly identified the pedestrian crossing sign (83.4%), followed by the no entry sign (76.8%) and roundabout ahead sign (75.8%). However, knowledge was particularly low for the school zone ahead sign, with only 14.9% of respondents identifying it correctly. These findings suggest that while riders are familiar with common regulatory and warning signs, knowledge of certain safety-related signs remains limited.

Table 2 Knowledge of road signs among respondents

Rank	Road Sign	Correct (%)
1	Pedestrian Crossing	83.4
2	No Entry	76.8
3	Roundabout Ahead	75.8
4	Give Way	71.8
5	No Overtaking	71.6
6	Speed Limit (50 km/h)	65.6
7	Slippery Road	63.3
8	Sharp Bend Right	58.3
9	Two-Way Traffic	55.7
10	School Zone Ahead	14.9

4.1.5 Overall knowledge of road signs by districts

Figure 3 presents respondents' knowledge of road safety signs across the selected districts. Respondents from Yilo Krobo recorded the highest knowledge level (49.3%), while those from Ablekuma Central had the lowest knowledge level (40.8%).

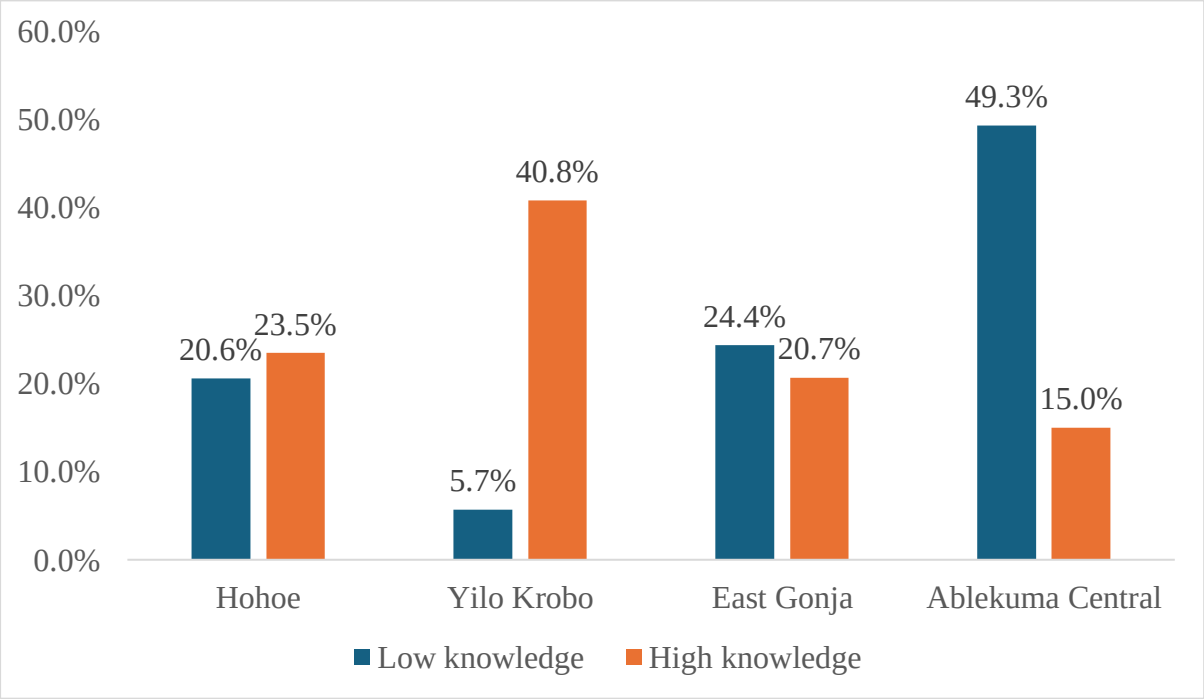

Figure 3 Overall knowledge of road signs by districts

4.1.6 Overall knowledge of road signs

Figure 4.3 shows the distribution of respondents according to their knowledge of road signs. Slightly more than half of the respondents (50.5%) demonstrated high knowledge of road signs, while 49.5% had low knowledge.

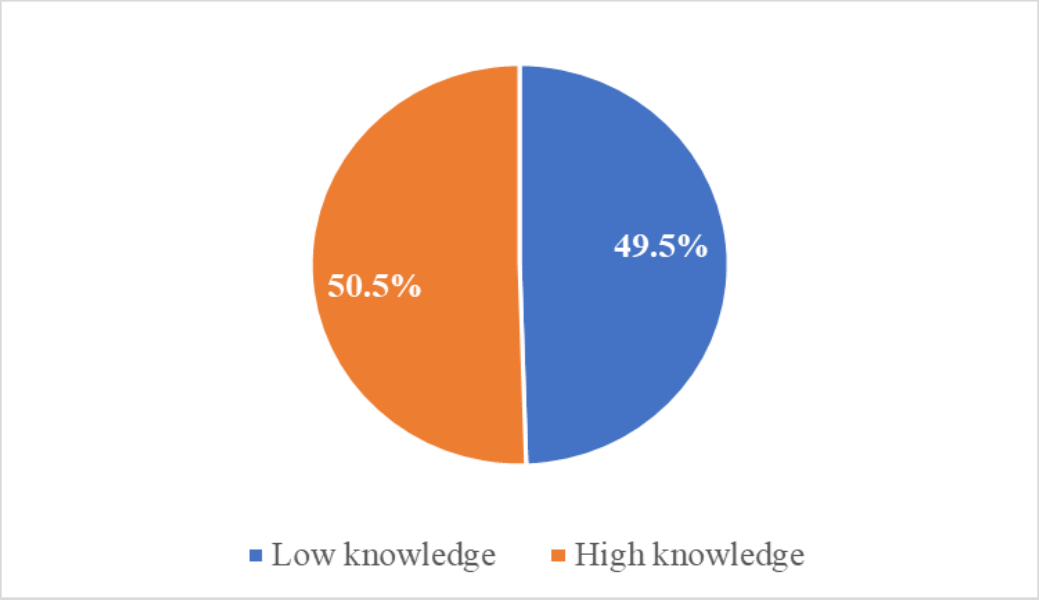

Figure 4 Overall knowledge of road signs

4.1.7 Demographic and behavioural factors associated with motorcycle accidents

The table shows a multivariable logistic regression analysis of the factors associated with involvement in a motorcycle accident. In the adjusted logistic regression model, stunt riding behaviour was significantly associated with accident involvement (aOR = 1.07, 95% CI: 1.02–1.11, p = 0.004). Riders in East Gonja were more likely to report accidents compared with those in Ablekuma Central (aOR = 2.46, 95% CI: 1.06–5.71), whereas riders in Yilo Krobo had significantly lower odds of accident involvement (aOR = 0.15, 95% CI: 0.05–0.51). Riders identifying as Muslims also had higher odds of accident involvement compared with Christians (aOR = 1.77, 95% CI: 1.01–3.09). Riding errors demonstrated a marginal association with accident involvement (p = 0.052), while other demographic and riding characteristics were not statistically significant predictors

Table 3 Demographic and behavioural factors associated with motorcycle accidents

Variable	cOR (95% CI), p-value	aOR (95% CI), p-value
Sex		
Male	Ref	
Female	0.86 (0.41-1.79), 0.678	0.78 (0.31-1.97), 0.601
Marital status		
Not in union	Ref	
In union	1.08 (0.68-1.71), 0.738	1.32 (0.69-2.52), 0.399
Age group		
Below 25	Ref	
25-34	0.81 (0.42-1.56), 0.532	0.81 (0.37-1.76), 0.591
35-44	1.30 (0.64-2.67), 0.462	1.02 (0.38-2.73), 0.975
45-54	1.15 (0.52-2.56), 0.733	0.96 (0.33-2.82), 0.939
55+	0.47 (0.09-2.28), 0.349	0.31 (0.05-1.84), 0.198
Religion		
Christianity	Ref	
Islam	**2.22 (1.38-3.58), 0.001**	**1.77 (1.01-3.09), 0.045**
African traditional	3.55 (0.95-13.25), 0.059	3.10 (0.69-14.02), 0.141
Educational attainment		
No formal educational	Ref	

Basic education	0.59 (0.27-1.34), 0.212	0.48 (0.18-1.32), 0.155
Secondary education	0.66 (0.33-1.31), 0.234	0.52 (0.22-1.21), 0.129
Tertiary	**0.37 (0.18-0.77), 0.008**	0.40 (0.16-1.01), 0.053
District		
Ablekuma central	Ref	
Yilo Krobo	**0.20 (0.09-0.45), <0.001**	**0.15 (0.05-0.51), 0.002**
East Gonja	1.46 (0.84-2.54), 0.177	**2.46 (1.06-5.71), 0.037**
Hohoe	**0.28 (0.13-0.59), 0.001**	0.37 (0.13-1.09), 0.071
Motorcycle license		
No	Ref	
Yes	0.82 (0.45-1.47), 0.499	0.68 (0.33-1.39), 0.298
Knowledge of road signs		
Low knowledge	Ref	
High knowledge	0.67 (0.42-1.06), 0.084	1.52 (0.79-2.92), 0.210
Riding purpose		
Commercial	Ref	
Non-commercial	1.11 (0.69-1.77), 0653	0.95 (0.46-1.96), 0.898
Motor ownership		
Yes	Ref	
No	1.01 (0.64-1.61), 0.955	1.15 (0.64-2.06), 0.650
Years of riding		

0-1	Ref	
1-5	1.43 (0.59-3.46), 0.430	0.96 (0.35-2.62), 0.936
Above 5	**2.44 (1.04-5.75), 0.041**	1.00 (0.34-3.01), 0.989
Riding frequency		
Every day	Ref	
Several times a week	1.51 (0.92-2.47), 0.100	0.75 (0.39-1.44), 0.384
Once a week	0.94 (0.26-3.46), 0.931	1.37 (0.27-7.00), 0.705
Rarely	0.38 (0.05-3.01), 0.358	0.45 (0.05-4.17), 0.481
Errors	1.01 (0.97-1.05), 0.752	0.93 (0.87-1.00), 0.052
Protective gear	0.97 (0.92-1.03), 0.305	0.96 (0.91-1.00), 0.088
Stunts	**1.08 (1.04-1.12), <0.001**	**1.07 (1.02-1.11), 0.004**
Speed violation	1.01 (0.98-1.04), 0.563	0.98 (0.89-1.07), 0.626

4.2 DISCUSSION

This study investigated the determinants of motorcycle crash involvement among riders in four selected municipalities in Ghana. The findings reveal a complex interplay between geographical location, religious affiliation, and specific riding behaviors in predicting the likelihood of accidents.

Prevalence of Accidents and Geographical Variations

The study found significant variations in crash prevalence across the four municipalities. **Ablekuma Central** reported a high prevalence of accidents, while the multivariable logistic regression showed that riders in **East Gonja** were significantly more likely to report accidents (aOR = 2.46, p = 0.037) compared to those in Ablekuma Central. Conversely, riders in **Yilo Krobo** had significantly lower odds of accident involvement (aOR = 0.15, p = 0.002) (Abia et al., 2019). These variations may be attributed to differences in road infrastructure and traffic density across urban and rural contexts. High traffic density in urban areas like Ablekuma often increases collision exposure, while rural settings like East Gonja may suffer from poor road surfaces and lack of lighting, which are known environmental risk factors in Ghana.

Behavioral Determinants: Stunt Riding and Errors

A key finding of this study was the significant association between **stunt riding behavior** and accident involvement (aOR = 1.07, p = 0.004). Dapilah et al. (2017) suggested that motorcyclists, particularly younger ones, often display higher risk-taking behaviors (such as speeding and risky overtaking) which contribute to higher crash rates. Interestingly, while general **riding errors** showed a marginal association with accident involvement (p = 0.052), it did not reach the standard

threshold for statistical significance. This suggests that intentional risky behaviors (stunts) may be more critical predictors of crashes in these districts than unintentional lapses in skill or judgment.

Socio-Demographic and Cultural Factors

The analysis revealed that **religious affiliation** was a significant predictor, with Muslim riders having higher odds of accident involvement compared to Christians (aOR = 1.77, p = 0.045) (Adanu et al., 2022). This may reflect underlying socio-economic or cultural differences in riding patterns or geographic location not fully captured by other variables.

Surprisingly, several factors traditionally linked to crash risk (**age**, **educational attainment**, and **years of riding**) did not reach statistical significance in the adjusted model (Abia et al. 2019). Konlan et al., (2022) highlights younger age as a primary risk factor due to inexperience, this study suggests that within these specific municipalities, behavioral choices like stunt riding and the environmental context of the district may be more dominant drivers of accident risk.

Knowledge of Road Signs

The study found that **knowledge of road signs was not a statistically significant predictor** of accident involvement in the multivariable model (aOR = 1.52, p = 0.210). This finding is noteworthy as it suggests that while riders may recognize and understand signs, this knowledge does not necessarily translate into safer riding outcomes. This gap between knowledge and practice is consistent with findings in other African contexts where riders may be aware of traffic signs but admit to not respecting them due to lack of patience or economic pressure. Furthermore, literature suggests that awareness of signs does not always equate to correct interpretation or behavioral compliance (Abia et al. 2019, Adanu et al., 2022).

Finally, the findings shows that interventions should move beyond general safety education and focus on district-specific risks and the discouragement of high-risk behaviors like stunt riding. The higher risk in areas like East Gonja highlights a need for improved road infrastructure and more stringent enforcement of traffic regulations in high-risk zones.

CHAPTER FIVE

5.0 CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

This study has provided critical insights into the factors predisposing commercial motorcycle riders to accidents across four distinct municipalities in Ghana. The findings demonstrate that motorcycle crash involvement is not a random occurrence but is significantly influenced by a combination of geographical, religious, and behavioral determinants.

Notably, the study found that riders in **East Gonja** are at a significantly higher risk of accidents, likely due to a combination of underdeveloped road infrastructure and environmental hazards common in rural settings. Conversely, the lower prevalence in **Yilo Krobo** suggests that certain local conditions or rider practices in that municipality may serve as protective factors. Behaviorally, **stunt riding** emerged as a powerful and statistically significant predictor of crashes, indicating that intentional risk-taking is a major driver of road traffic injuries among this population.

Surprisingly, while half of the riders demonstrated high knowledge of road signs, this knowledge did not translate into reduced accident involvement. This confirms a "knowledge-practice gap," where cognitive awareness of safety rules is often overridden by economic pressures or behavioral tendencies. In summary, while socio-demographic factors like age and education are important, the geographical context of riding and the specific engagement in risky behaviors like stunts are the most critical predictors of motorcycle accidents in the study areas.

5.2 Recommendations

Based on the findings, the following recommendations are proposed to enhance motorcycle safety and reduce crash involvement:

We recommend district-specific safety interventions: Authorities should move away from "one-size-fits-all" policies and instead address the unique risks found in each area, such as focusing on rural road hazards in East Gonja versus urban traffic density in Ablekuma Central.

We recommend stricter enforcement against high-risk behaviors: Law enforcement agencies (MTTD) should prioritize monitoring and penalizing stunt riding, as our data identifies this as a primary and significant predictor of accidents.

We recommend transitioning to cognitive-behavioral training: Since knowledge of signs alone does not prevent crashes, we recommend that rider education programs focus on behavioral change and the practical application of safety rules under economic pressure.

We recommend targeted infrastructure audits: We recommend that the Ministry of Roads and Highways conducts safety audits in the high-risk municipalities identified in this study to fix unpaved surfaces and improve drainage and lighting.

We recommend a "Bottom-Up" approach to road signage: To bridge the knowledge-practice gap, we recommend involving local rider unions in the placement and education of road signs to ensure they are respected and understood within the local context.

We recommend regularized motorcycle maintenance checks: To reduce mechanical failure as a crash factor, we recommend mandatory, periodic roadworthiness inspections specifically for commercial "Okada" motorcycles.

We recommend further investigation into socio-cultural determinants: We recommend that future researchers explore the underlying reasons why specific demographic groups, such as those identified by religious affiliation in our study, show different odds of accident involvement.

REFERENCES

Abia, F., Essono, M., & Etoundi, M. (2019). Knowledge of motorcycle riders on road safety and traffic signs in Yaoundé, Cameroon. *Journal of Public Health in Developing Countries, 6*(2), 60-66.

Adanu, E. K., Aidoo, E. N., & Mohammed, A. S. (2022). Motorcycle crash fatalities and injuries in Ghana: Patterns and trends. *World Journal of Advanced Research and Reviews, 15*(2), 294-302.

Adu-Gyamfi, E. (2022). Fatalism and risk perception among commercial motorcycle riders in Eastern Ghana. *Ghana Journal of Social Sciences, 19*(2), 45–58.

Ahmed, S. K., Mohammed, M. G., Abdulqadir, S. O., El-Kader, R. G. A., El-Shall, N. A., Chandran, D., ... & Dhama, K. (2023). Road traffic accidental injuries and deaths: A neglected global health issue. *Health science reports*, *6*(5), e1240.

Aidoo, E. N., Amoh, S. E., & Agyei, S. K. (2021). Injury patterns among motorcycle crash victims in Ghana. *BMC Public Health, 21*(1), 112–120. https://doi.org/10.1186/s12889-021-11210-1

Alhassan, R. K., Adongo, P. B., & Mock, C. N. (2022). Risk factors for head injuries among motorcycle crash survivors in northern Ghana. *BMC Public Health, 22*(1), 2050.

Appiah, A. B., Akweongo, P., Sackey, S., Morna, M. T., Kenu, P., Dun-Boibuo, A., Donkor, P., & Mock, C. N. (2022). Factors associated with head injury among survivors of motorcycle crashes: a case–control study in northern Ghana. *Pan African Medical Journal.* https://www.panafrican-med-journal.com/content/article/43/73/full

Appiah, S., Yakubu, I., & Fuseini, A. (2021). Risk perception and accident attribution among motorcycle riders in Northern Ghana. *West African Journal of Transportation and Safety, 3*(1), 55–67.

Asamoah, P. K. B., Kombonaah, N., & Appiah Atiemo, E. (2019). Understanding why motorcycle riders do not comply with traffic control signals in the Wa Municipality in Ghana. *Research on Humanities and Social Sciences, 9*(14), 31-40. https://doi.org/10.7176/RHSS/9-14-17

Bloomberg Philanthropies Initiative for Global Road Safety. (2022). *Road safety in low- and middle-income countries: Policy and infrastructure improvements.* Bloomberg Philanthropies.

Cochran, W. G. (1977). *Sampling techniques* (3rd ed.). John Wiley & Sons.

Creswell, J. W., & Creswell, J. D. (2018). *Research design: Qualitative, quantitative, and mixed methods approaches* (5th ed.). SAGE Publications.

Dapilah, F., Guba, B., & Adongo, R. (2017). Motorcyclist characteristics and traffic behaviour in urban northern Ghana: Implications for road traffic accidents. *Transportation Research Procedia, 25,* 1788–1797. https://doi.org/10.1016/j.trpro.2017.05.189

Dapilah, F., Guba, B., & others. (2017). Motorcyclist characteristics and traffic behaviour in urban Northern Ghana: implications for road traffic accidents. *Transportation Research Procedia.* https://www.sciencedirect.com/science/article/abs/pii/S2214140516000232

FIA Foundation. (2022, November 16). *Sub-Saharan motorcycle boom puts lives at risk.* https://www.fiafoundation.org/news/sub-saharan-motorcycle-boom-puts-lives-at-risk-warns-new-fia-foundation-report

Ghana News Agency (GNA). (2022, December 12). Motorcycle fatalities constitute 34 per cent of Ghana's road-traffic crashes in 2021 – NRSA. *Ghana News Agency.* https://www.gna.org.gh/

Ghana News Agency. (2024, August 15). Dusty Huhunya road kills one, 14 in critical condition. *Ghana News Agency.*

Ghana Police Motor Traffic and Transport Department (MTTD). (2022). *Annual road crash statistical report.* Accra: MTTD Headquarters.

Global Road Safety Facility (GRSF). (2023). *Motorcycle safety in Africa.* World Bank. https://www.globalroadsafetyfacility.org/

Global Road Safety Facility. (2023). *Motorcycle safety in Africa.* World Bank. https://www.globalroadsafetyfacility.org/sites/default/files/2023-10/Motorcycle%20Safety%20in%20Africa.pdf

Graphic Online. (2024, May 22). Crushing okada crashes: Helmet-use campaign intensifies. *Graphic Online.* https://www.graphic.com.gh/news/general-news/crushing-okada-crashes-helmet-use-campaign-intensifies.html

Hagan, D., Tarkang, E. E., & Aku, F. Y. (2021). Compliance of commercial motorcycle riders with road safety regulations in a peri-urban town of Ghana. *PLOS ONE, 16*(3), e0246965. https://doi.org/10.1371/journal.pone.0246965

Hassan, M., & Yeboah, P. (2021). Religion, fate, and perceptions of road safety among Ghanaian road users. *Journal of Religion and Society, 23*(4), 97–110.

Independent. (2022, August 23). Improving riders' knowledge of road safety signs, markings. *Independent (Nigeria).* Retrieved from https://independent.ng/improving-riders-knowledge-of-road-safety-signs-markings/

Konlan, K. D., Abekah-Carter, K., & Amegah, K. E. (2022). Factors associated with motorcycle-related road traffic crashes in Africa: A scoping review. *BMC Public Health, 22*(1), 2050. https://doi.org/10.1186/s12889-022-13075-2

Konlan, K. D., et al. (2022). Factors associated with motorcycle-related road traffic crashes in Africa: a scoping/systematic review (2016–2022). *BMC Public Health, 22.* https://doi.org/10.1186/s12889-022-13075-2

Lajunen, T., & Özkan, T. (2011). Attributional factors and drivers' self-assessment of accident risk. *Transportation Research Part F: Traffic Psychology and Behaviour, 14*(6), 445–453. https://doi.org/10.1016/j.trf.2011.08.001

Levin, K. A. (2006). Study design III: Cross-sectional studies. *Evidence-Based Dentistry, 7*(1), 24–25. https://doi.org/10.1038/sj.ebd.6400375

Levy, P. S., & Lemeshow, S. (2013). *Sampling of populations: Methods and applications* (4th ed.). John Wiley & Sons.

Maulina, D., Andini, E., Tanri, D., Kamsanih, N., Mazaya, S., & Zhafirah, T. (2022). Traffic signs comprehension among motorcyclists: The effect of sign type and sign familiarity. *Humaniora, 13*(1), 81-90. https://doi.org/10.21512/humaniora.v13i1.7379

Miller, D. T., & Ross, M. (1975). Self-serving biases in the attribution of causality: Fact or fiction? *Psychological Bulletin, 82*(2), 213–225. https://doi.org/10.1037/h0076486

Ministry of Food and Agriculture (MoFA). (2024). *Hohoe Municipal: District agricultural profile.* MoFA Ghana. https://mofa.gov.gh/site/sports/district-directorates/volta-region/281-hohoe-municipal

Mock, C. N., Amegashie, J., & Dartey, S. (2012). Economic cost of road-traffic crashes in northern Ghana. *Injury Prevention, 18*(3), 125–131. https://doi.org/10.1136/injuryprev-2011-040151

Mustapha, S. D., & Ibitoye, B. A. (2022). Understanding of traffic signs by drivers on urban roads: A case study of Ilorin, Kwara State. *Journal of Engineering Research and Reports, 23*(12), 39-47. https://doi.org/10.9734/JERR/2022/v23i12761

MyJoyOnline. (2024, August 14). Dusty Huhunya road kills one, 14 in critical condition. *MyJoyOnline.* https://www.myjoyonline.com/dusty-huhunya-road-kills-one-14-in-critical-condition

Nguyen, H. L., & Ha, T. M. (2019). Perceived causes of motorcycle crashes and safety behaviours among Vietnamese riders. *International Journal of Injury Control and Safety Promotion, 26*(3), 228–237. https://doi.org/10.1080/17457300.2019.1597837

Oduro-Appiah, K., Osei, P., & Ntiamoah, K. (2020). Infrastructure and crash attribution along major highways in Ghana. *African Transport Policy Journal, 7*(1), 33–49.

Oladepo, O., & Brieger, W. R. (2020). Attribution of road crashes among commercial motorcyclists in southwestern Nigeria. *Journal of Safety Research, 73*(5), 131–139. https://doi.org/10.1016/j.jsr.2020.03.006

Owusu-Ansah, P., Abdul-Aziz, A. R., Obiri-Yeboah, A. A., Amoah, A. S., Woangbah, S. K., & Adusei, E. (2024). Modelling road fatalities from tricycle crashes in Ashanti Region, Ghana: An application of regression with ARIMA errors. *Transportation Research Interdisciplinary Perspectives*, *26*, 101180.

Owusu, S. A., Darko, E. A., & Boateng, A. S. (2024). Attitudes, intentions and helmet use among motorcyclists in Ghana. *International Journal of Injury Control and Safety Promotion, 31*(1), 55-64.

Polit, D. F., & Beck, C. T. (2021). *Nursing research: Generating and assessing evidence for nursing practice* (11th ed.). Wolters Kluwer Health.

Prevalence and risk factors for motorcycle accidents presenting to the Korle-Bu Teaching Hospital, Ghana. (2022). *World Journal of Advanced Research and Reviews, 16*(1), 1048. https://doi.org/10.30574/wjarr.2022.16.1.1048

Reason, J. (1990). *Human error.* Cambridge University Press.

Road Traffic Management Corporation (RTMC). (2023). *African Region status report on road safety 2023.* Pretoria, South Africa.

Subasish, B., Ramli, R., & Wong, S. V. (2018). External versus internal attributions of road traffic crashes among Malaysian motorcyclists. *Asian Journal of Social Psychology, 21*(4), 250–262. https://doi.org/10.1111/ajsp.12310

The Ghana Report. (2024, May 17). Motorcyclists and pedestrians top casualties of road crash deaths. *The Ghana Report.* https://www.theghanareport.com/motorcyclists-pedestrians-top-casualties-of-road-crash-deaths

Weiner, B. (1985). An attributional theory of achievement motivation and emotion. *Psychological Review, 92*(4), 548–573. https://doi.org/10.1037/0033-295X.92.4.548

Wikipedia. (2024). *East Gonja Municipal District.* In *Wikipedia: The Free Encyclopedia.* https://en.wikipedia.org/wiki/East_Gonja_Municipal_District

World Health Organization (WHO). (2023). *Road traffic injuries: Fact sheet.* Geneva: WHO. https://www.who.int/news-room/fact-sheets/detail/road-traffic-injuries

World Health Organization African Region. (2024). *Status report on road safety in the WHO African Region 2023.* Brazzaville: WHO Regional Office for Africa.

World Health Organization Regional Office for Africa. (2023). *Status report on road safety in the WHO African Region, 2023.* https://www.afro.who.int/publications/road-safety-who-african-region-2023

World Health Organization. (2023). *Road traffic injuries: Fact sheet.* https://www.who.int/news-room/fact-sheets/detail/road-traffic-injuries

Zhou J, Zheng T, Dong S, Mao X, Ma C. Impact of Helmet-Wearing Policy on E-Bike Safety Riding Behavior: A Bivariate Ordered Probit Analysis in Ningbo, China. International Journal of Environmental Research and Public Health. 2022; 19(5):2830. https://doi.org/10.3390/ijerph19052830

APPENDICES

Appendix I: Work Plan

ACTIVITY	PERIOD					
	September, 2025-November, 2025	February, 2026	March, 2026	April, 2026	April, 2026	May, 2026
Proposal Development and submission						
Submission of proposal to the ethical review committee						
Recruitment and training of field officers						
Data collection						
Data Analysis						
Final write-up for Submission						

Appendix II: Budget

Activity	Resource	Quantity	Number of people	frequency	Unit price	Total
A. Proposal development	A4 papers	1 Rim		1	70.00	70.00
	Comb binding	5 Pairs		1	5.00	25.00
	Printing	1		1	250.00	250.00
	Lamination	5 Pairs		1	3.00	15.00
Total for A						360.00
B. Pre-testing of questionnaires	A4 pairs					
	Transportation to site		1	1	50.00	50.00
Subtotal for B						50.00
C. Training of data collectors	Refreshment	1 serving	5	1	20	100.00
	Transportation on site		5	1	20	100.00
Subtotal C						200.00
D. Data collection	Transportation on site		5	2	20	200 .00
	A4 sheet	3 Rims		2	70	420.00
	Lunch for data collectors		5	2	30	300.00
Subtotal D						920.00
E. Data entry and analysis	Lunch for data Entry	1				100.00
Contingency						200.00
Subtotal E						300.00
Grand total(A+B+C+D+E)						2000.00

Appendix III: Informed Consent Form

RESEARCH OPERATIONS OFFICE

INSTITUTE OF HEALTH RESEARCH

UNIVERSITY OF HEALTH AND ALLIED SCIENCES

RESEARCH ETHICS COMMITTEE (REC)

PROTOCOL CONSENT FORM

Section A- **BACKGROUND INFORMATION**

Title of Study:	**Determinants of Motorcycle Crash Involvement Among Commercial Riders in Selected Municipalities of Ghana.**
Principal Investigators:	Mr. Amanya Kennedy Nanor**,** Mr. Nelson Apala Atambire, Mr. Emmanuel Adu-Awuku, Miss Philomina Amo Appiah
Certified Protocol Number	

Section B– **CONSENT TO PARTICIPATE IN RESEARCH**

General Information about Research

This form describes a research project. It has information to help you decide whether you wish to participate. Research studies include only people who choose to take part, your participation is completely voluntary. Please discuss any questions you have about the study or about this form with the project staff before deciding to participate.

The study is being conducted by Mr. Amanya Kennedy Nanor**,** Mr. Nelson Apala Atambire, Mr. Emmanuel Adu-Awuku, Miss Philomina Amo Appiah, students of the University of Health and Allied Sciences (UHAS). You are being asked to be part of this study as a participant who engage in motor riding. The aim of the study is **to identify and assess the predisposing risk factors associated with motorcycle accidents among riders in selected districts of Ghana.**

Benefits/Risks of the study

The study will not cause any form of risk or harm to you, apart from the time you will spend during this interview, your risk may include answering sensitive questions concerning your age, education and other socio-demographic factors. There are also no individual benefits

Confidentiality

Your privacy and anonymity are assured and information collected from you will be treated as confidential. All data obtained will be handled anonymously and confidentially. Data will be stored in well-secured cabinets and only accessible to the principal investigators. Your name will not be required in any way to run the analysis.

However, in case your name is being captured, the information that will be provided by you will be kept private from access by anyone not involved in the study.

Compensation

There will be no compensation for participating in the study. But you will also be duly acknowledged in the final thesis for your participation, cooperation and contribution.

Withdrawal from Study

You will have the liberty to withdraw at any stage when you do not wish to continue. If you feel distressed or uncomfortable with some of the questions or do not want to answer some questions, you may choose not to answer them. What happens after study or when the participant changes his/her mind? A participant who decides to withdraw from the study or change his/her mind about the study will not be stopped or prevented. Accordingly, information collected from participants who decide to withdraw from the study will not be added to the final data for analysis.

Contact for Additional Information

If you have any question or any relevant issues about the study, you may contact the:

PRINCEPAL INVESTIGATORS: Mr. Amanya Kennedy Nanor, 0551839928, kennedyamanya2020@gmail.com, Mr. Nelson Apala Atambire, 0592991230, nelsonatambire@gmail.com, Mr. Emmanuel Adu-Awuku, 0556925733, aduawuku.jr@gmail.com, Miss Philomina Amo Appiah, 0547298833, appiahphilomina54@gmail.com.

SUPERVISOR: Dr. Mawuli Kushitor, gmail: mkushitor@uhas.edu.gh

University of Health and Allied Sciences (Fred N. Binka School of Public Health)

If you have any questions about your rights as a research participant in this study you may contact the Administrator of the Research Ethics Committee, IHR, University of Health and Allied Sciences at **rec@uhas.edu.gh or +233- 362-196-193**.

Section C- **PARTICIPANT AGREEMENT**

"I have read or have had someone read all of the above, asked questions, received answers regarding participation in this study, and am willing to give consent for me, my child/ward to participate in this study. I will not have waived any of my rights by signing this consent form. Upon signing this consent form, I will receive a copy for my personal records."

_____

Name of Participant

_____ _____

Signature or mark of Participant Date

If participant cannot read and or understand the form themselves, a witness must sign here:

I was present while the benefits, risks and procedures were read to the volunteer. All questions were answered and the volunteer has agreed to take part in the research.

_____

Name of witness

_____ _____

Signature of witness / Mark Date

I certify that the nature and purpose, the potential benefits, and possible risks associated with participating in this research have been explained to the above individual.

_____

Name of Person who Obtained Consent

_____ _____

Signature of Person Who Obtained Consent Date

Appendix IV: Data Collection Instrument

UNIVERSITY OF HEALTH AND ALLIED SCIENCES, SCHOOL OF PUBLIC HEALTH

TOPIC: DETERMINANTS OF MOTORCYCLE CRASH INVOLVEMENT AMONG COMMERCIAL RIDERS IN SELECTED MUNICIPALITIES OF GHANA.

REQUEST FOR YOUR PARTICIPATION

I am ……. a final year student of the University of Health and Allied Sciences and I am undertaking a study on the above topic as partial fulfilment of the requirement for the award of a degree in BSc. Public Health.

This research is purely an academic exercise and has been structured such that your contribution will be kept anonymous and confidential. You have the right to withdraw from the study at any given period without suffering any intimidation but your contribution is necessary to the success of this study.

Counting on your co-operation.

Thank you.

The original motor riders behavior questionnaire (MRBQ) **Developed by Elliott, T., Baughan, C., & Sexton, B. (2007)** has 43 items, and for each item, the respondents are asked to rate the frequency of their riding behaviour during last year by choosing one option from the 6-point scale: 1=never, 2=hardly ever, 3=occasionally, 4=quite often, 5=frequently, and 6=nearly all the time.

Sociodemographic	
Age	
District	1 Yilo Krobo 2 Ablekuma Central 3 Hohoe 4 East Gonja
Riding years	

Do you have a valid motorcycle driving license?	Yes/no/applying/expired
License years	
Marital status	Married Not married
No of dependents	
Gender	Male Female
Highest education	Primary SHS Tertiary
Holding a driver's license	Yes No
Riding frequency	Everyday Several times per week Once a week Less than once a week
Average trip per frequency	An integer
Main riding purpose	Carry for free Carry for money Others Relax/travel/sports To work/study place
Average monthly income (carry for money only)	
Mileage: Average annual mileage (km)	
Having own motorcycle	No yes
Experience near crash in the past 12 months	No yes

Experience crash in the past 12 months	No yes
Experience near crash in the past 1 months	No yes
Experience crash in the past 1 months	No yes
Experience near crash in the past 1 week	No yes
Experience crash in the past 1 week	No yes
Offences: Number of penalized traffic violations (last 12 months)	No yes
Motor riders behavior questionnaire MRBQ	
1. Fail to notice that pedestrians are crossing when turning into a side street from a main road	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time.
2. Not notice someone stepping out from behind a parked vehicle until it is nearly too late	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
3. Not notice a pedestrian waiting at a crossing where the lights have just turned red	1=never 2=hardly ever 3=occasionally 4=quite often

	5=frequently 6=nearly all the time
4. Pull onto a main road in front of a vehicle you have not noticed or whose speed you misjudged	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
5. Miss "Give Way" signs and narrowly avoid colliding with traffic having right of way	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
6. Fail to notice or anticipate another vehicle pulling out in front of you and had difficulty stopping	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
7. Queuing to turn left (in England; turn right in other countries) on a main road, you pay such close attention to the mainstream of traffic that you nearly hit the car in front	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
8. Distracted or pre-occupied, you suddenly realize that the vehicle in front has slowed, and you have to brake hard to avoid a collision	1=never 2=hardly ever 3=occasionally

	4=quite often 5=frequently 6=nearly all the time
9. Attempt to overtake someone that you had not noticed to be signaling a right turn (in England; left turn in other countries)	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
10. When riding at the same speed as other traffic, you find it difficult to stop in time when a traffic light has turned against you	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
11. Ride so close to the vehicle in front that it would be difficult to stop in an emergency	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
12. Run wide when going around a corner	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
13. Ride so fast into a corner that you feel like you might lose control	1=never 2=hardly ever

	3=occasionally 4=quite often 5=frequently 6=nearly all the time
14. Exceed the speed limit on a country/rural road	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
15. Disregard the speed limit late at night or in the early hours of the morning	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
16. Disregard the speed limit on a motorway	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
17. Disregard the speed limit on a residential road	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time

18. Race away from traffic lights with the intention of beating the driver next to you	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
19. Open up the throttle and just go for it on a country road	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
20. Ride between two lanes of fast moving traffic	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
21. Got involved in racing other riders or drivers	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
22. Ride so fast into a corner that you scare yourself	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently

	6=nearly all the time
23. Attempt or done a wheelie	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
24. Pull away too quickly and your front wheel lifted off the road	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
25. Intentionally do a wheel spin	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
26. Unintentionally do a wheel spin	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
27. Wear motorcycle riding boots	1=never 2=hardly ever 3=occasionally 4=quite often

	5=frequently 6=nearly all the time
28. Wear protective trousers– leather or non-leather	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
29. Wear a protective jacket– leather or non-leather	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
30. Wear body armour/impact protection for the elbows and shoulders	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
31. Wear no protecting clothing	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
32. Wear motorcycle gloves	1=never 2=hardly ever 3=occasionally

	4=quite often 5=frequently 6=nearly all the time
33. Wear bright fluorescent strips/patches on your clothing	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
34. Use daytime headlights on your bike	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
35. Brake or throttle back (slow down) when going around a bend	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
36. Change gears when going round a corner or bend	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
37. Find that you have difficulty controlling the bike when riding at speed (e.g. steering wooble)	1=never 2=hardly ever

	3=occasionally 4=quite often 5=frequently 6=nearly all the time
38. Skid on a wet road or manhole cover, road making	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
39. Have trouble with your visor or goggles fogging up	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
40. Another driver deliberately annoys you or puts you at risk	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
41. Ride when you suspect you might be over the legal limit for alcohol	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
42. Wear a full leather-suit	1=never

	2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time
43. Wear bright/fluorescent clothing	1=never 2=hardly ever 3=occasionally 4=quite often 5=frequently 6=nearly all the time

ROAD SIGN KNOWLEDGE TEST

For each of the following road signs, look at the picture and select the correct meaning.

Item 1: (Insert Picture: "No Entry")

What does this road sign mean?

1. No parking
2. No entry **(Correct)**
3. No overtaking
4. Give way

Item 2: (Insert Picture: "Sharp Bend to the Right")

What does this sign indicate?

1. Two-way traffic
2. Road narrows
3. Sharp right bend ahead **(Correct)**

4. Slippery road

Item 3: (Insert Picture: "Pedestrian Crossing")

This sign means:

1. School ahead
2. Pedestrian crossing **(Correct)**
3. Stop and wait
4. Children at play

Item 4: (Insert Picture: "Give Way")

What does this sign mean?

1. Stop completely
2. Give way / Yield **(Correct)**
3. No motorcycles
4. Traffic signals ahead

Item 5: (Insert Picture: "Roundabout Ahead")

This sign indicates:

1. No left turn
2. Roundabout ahead **(Correct)**
3. U-turn allowed
4. Keep right

Item 6: (Insert Picture: "No Overtaking")

This sign means:

1. No speeding
2. No overtaking **(Correct)**

3. No honking
4. No stopping

Item 7: (Insert Picture: "Slippery Road")

This warning sign shows:

1. Rough road ahead
2. Slippery road **(Correct)**
3. Steep hill
4. Road surface uneven

Item 8: (Insert Picture: "School Zone Ahead")

This sign means:

1. Children crossing
2. School zone ahead **(Correct)**
3. Pedestrian crossing
4. Market area

Item 9: (Insert Picture: "Speed Limit – 50 km/h")

This sign means:

1. Minimum speed is 50
2. Maximum speed is 50 **(Correct)**
3. Recommended speed is 50
4. No speed limit

Item 10: (Insert Picture: "Two-Way Traffic Ahead")

This sign informs you that:

1. One-way road

2. Dual carriageway
3. Two-way traffic ahead **(Correct)**
4. No vehicles allowed